# Potentiation of gemcitabine by Turmeric Force in pancreatic cancer cell lines.

Gemcitabine is a first line cancer drug widely used for the treatment of pancreatic cancer. However, its therapeutic efficiency is significantly limited by resistance of pancreatic cancer cells to this and other chemotherapeutic drugs. We have investigated the cytotoxic effect of Turmeric Force (TF), a supercritical and hydroethanolic extract of turmeric, alone and in combination with gemcitabine in two pancreatic carcinoma cell lines (BxPC3 and Panc-1). TF is highly cytotoxic to BxPC3 and Panc-1 cell lines with IC50 values of 1.0 and 1.22 microg/ml, respectively with superior cytotoxicity than curcumin. Gemcitabine IC50 value for both of these cell line is 0.03 microg/ml; however, 30-48% of the pancreatic cancer cells are resistant to gemcitabine even at concentrations >100 microg/ml. In comparison, TF induced cell death in 96% of the cells at 50 microg/ml. The combination of gemcitabine and TF was synergistic with IC90 levels achieved in both pancreatic cancer cell lines at lower concentrations. CalcuSyn analysis of cytotoxicity data showed that the Gemcitabine + Turmeric Force combination has strong synergism with combination index (CI) values of 0.050 and 0.183 in BxPC3 and Panc-1 lines, respectively at IC50 level. This synergistic effect is due to the increased inhibitory effect of the combination on nuclear factor-kappaB activity and signal transducer and activator of transcription factor 3 expression as compared to the single agent.